

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 416 — Past Questions & Materials
by Renance

Course: MTH 416
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — MTH 416 2020 2021 Time Series First Semester

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS

END OF SEMESTER EXAMINATION FOR MTH 416 - (TIME SERIES) 2020/2021
INSTRUCTION: ANSWER ANY 4 QUESTIONS CA: 30 MARKS 2HRS 10MINS

1. (a) Define the following terms:
 - (i) Time Series (1 MARK)
 - (ii) Discrete time Series (1 MARK)
 - (iii) Continuous time series (1 MARK)
 (b) Briefly explain the following:
 - (i) Four major objective of a time series (4 MARKS)
 - (ii) Deseasonalised series (3 MARKS)
 (c) A company management finds that the sales of its products are affected by certain period of the year which tends to obscure any trend. The following sales data are available showing sales in successive periods of four months

Period	2017	2018	2019	2020
Jan. - April	84	92	94	100
May - Aug.	121	140	148	161
Sept. - Dec.	67	81	83	95

- (i) Using the 3-moving average technique, estimate the trend and calculates seasonal effects (4.5 MARK)
- (ii) By using your trend in (i) above, estimate the mean monthly increase in sales that the company has achieved over the period. (3 MARKS)

2. (a) (i) What is correlogram? (1.5 MARK)
- (ii) What are the characteristics of the autocorrelation coefficients when we have an alternative series, random series and stationary series? (3 MARKS)
- (b) Distinguish between the following:
 - (i) Strict stationary and m-th order stationary (2 MARKS)
 - (ii) Maximum likelihood and Jacknife estimators of an autocovariance function (2 MARKS)
 - (iii) Time domain and frequency domain analysis. (2 MARKS)
- (c) Show that for a given time series $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$. The autocovariance generating function is $\Gamma(B) = \sigma^2 (\psi(B)\psi(B^{-1}))$. (7 MARKS)

3. (a) Show that for $|\rho(\tau)| \leq 1$ for all τ , $|\gamma(\tau)| \leq \gamma(0)$. (3 MARKS)
- (b) Define the following
 - (i) Seasonal variation (2 MARKS)
 - (ii) Moving average (2 MARK)
 - (iv) Additive model representation (2 MARK)

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